São Paulo Tech School

Gustavo Lima de Araujo

Docker pt.01

Evidências da instalação, configuração e utilização do Docker, instalando e utilizando um container a partir de uma imagem.

Rua Haddock Lobo, 595

10.2026

Gustavo Lima de Araujo

Docker pt.01

Evidências da instalação, configuração e utilização do Docker, instalando e utilizando um container a partir de uma imagem.

Este documento tem como objetivo estudar e enfatizar conceitos sobre a instalação e configuração do Docker, além de buscar uma imagem no registry e utilizar ela, colocando as informações do banco de dados do projeto dentro do container.

Orientador: Eduardo Verri

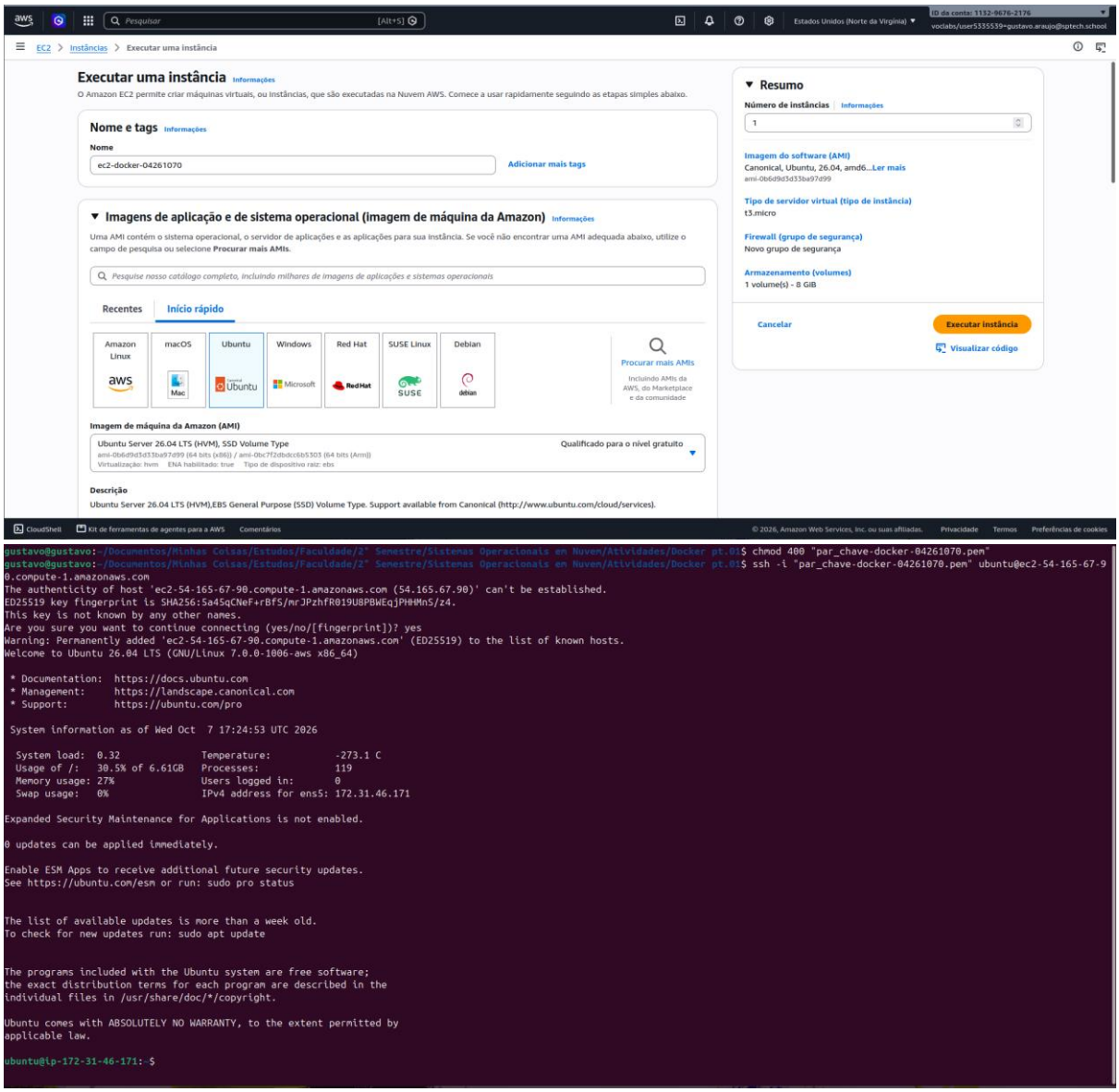
Rua Haddock Lobo, 595

10.2026

Sumário

1. Configurando e utilizando EC2 3
2. Instalação do Docker 4
3. Criando um container 5
4. Entrando no Bash do Container 6
5. Utilizando o banco de dados do Container 7
6. Conferindo informações do Banco de Dados 9
7. Desligando/Parando um Container 9
..... 9
8. Deletando um Container 9
9. Deletando uma Imagem 9

1. Configurando e utilizando EC2



2. Instalação do Docker

```
 GNU nano 8.7.1                                        install-docker.sh *
#!/bin/bash

# Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update

sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

Write to File: install-docker.sh
^G Help          M-D DOS Format      M-A Append          M-B Backup File     ^T Browse
^C Cancel        M-M Mac Format      M-P Prepend         ^O Discard buffer
```

```
ubuntu@ip-172-31-46-171:~$ nano install-docker.sh
ubuntu@ip-172-31-46-171:~$ chmod +x install-docker.sh
ubuntu@ip-172-31-46-171:~$
```

```
ubuntu@ip-172-31-46-171:~$ sudo systemctl start docker.service
ubuntu@ip-172-31-46-171:~$ sudo systemctl status docker.service
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-07 17:28:11 UTC; 1min 9s ago
 Invocation: ac3e4c2fa2824afeabcbc3788ef94ebd
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 7341 (dockerd)
      Tasks: 9
     Memory: 115.6M (peak: 115.6M)
        CPU: 640ms
     CGroup: /system.slice/docker.service
             └─7341 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 07 17:28:10 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:10.642049149Z" level=info msg="Restoring containers: start."
Oct 07 17:28:10 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:10.719507001Z" level=info msg="Deleting nftables IPv4 rules" error="running nft: /dev/stdin:1:17-30: Error: Could not
Oct 07 17:28:10 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:10.725653379Z" level=info msg="Deleting nftables IPv6 rules" error="running nft: /dev/stdin:1:18-31: Error: Could not
Oct 07 17:28:11 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:11.120290846Z" level=info msg="Loading containers: done."
Oct 07 17:28:11 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:11.141862886Z" level=info msg="Docker daemon" commit=8af9fe3 containerd-snapshotter=true storage-driver=overlayfs vers
Oct 07 17:28:11 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:11.142972775Z" level=info msg="Initializing buildkit"
Oct 07 17:28:11 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:11.532129400Z" level=info msg="Completed buildkit initialization"
Oct 07 17:28:11 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:11.542292788Z" level=info msg="Daemon has completed initialization"
Oct 07 17:28:11 ip-172-31-46-171 dockerd[7341]: time="2026-10-07T17:28:11.542552144Z" level=info msg="API listen on /run/docker.sock"
Oct 07 17:28:11 ip-172-31-46-171 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-23/23 (END)
```

```
ubuntu@ip-172-31-46-171:~$ sudo systemctl enable docker.service
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
ubuntu@ip-172-31-46-171:~$
```

```
ubuntu@ip-172-31-46-171:~$ sudo usermod -aG docker $USER
```

3. Criando um container

```
ubuntu@ip-172-31-46-171:~$ docker run -d -p 3306:3306 --name db -e "MYSQL_ROOT_PASSWORD=urubu100" mysql:8.0.44
Unable to find image 'mysql:8.0.44' locally
8.0.44: Pulling from library/mysql
afd8a12fa4e1: Pull complete
ea7ef84dcaea: Pull complete
658e67031dba: Pull complete
ef7bcbef3f90: Pull complete
f40e06a2cd88: Pull complete
caee3c6664fb: Pull complete
83bab8f9c85f: Pull complete
655eb0dfc8fe: Pull complete
80caa0f442af: Pull complete
c9ae3e76dfc9: Pull complete
cc3a161f487a: Pull complete
b622c0882d0e: Download complete
a2108fc02a60: Download complete
Digest: sha256:9c3380eac945af0736031b200027f581925927c81e010056214a4bd6b6693714
Status: Downloaded newer image for mysql:8.0.44
05a9df2f23c4f9116c8bb50a611f31b1fea380d2d979ddcfeb6ffdf71add13d6
ubuntu@ip-172-31-46-171:~$
```

```
ubuntu@ip-172-31-46-171:~$ docker images

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:8.0.44   9c3380eac945       1.08GB          247MB   U
ubuntu@ip-172-31-46-171:~$
```

```
ubuntu@ip-172-31-46-171:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                                                  NAMES
05a9df2f23c4   mysql:8.0.44   "docker-entrypoint.s…"   About a minute ago   Up About a minute   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   db
ubuntu@ip-172-31-46-171:~$
```

4. Entrando no Bash do Container

```
ubuntu@ip-172-31-46-171:~$ docker exec -it db bash
bash-5.1# mysql -u root -purubu100
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.44 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

5. Utilizando o banco de dados do Container

```
Query OK, 1 row affected (0.01 sec)

mysql> USE noctua_bd;
Database changed
mysql> CREATE TABLE empresa (
    ->     id_empresa INT PRIMARY KEY AUTO_INCREMENT,
    ->     razao_social VARCHAR(60) NOT NULL,
    ->     cnpj CHAR(14) NOT NULL UNIQUE,
    ->     dominio VARCHAR(60) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE permissao (
    -> id_permissao INT PRIMARY KEY AUTO_INCREMENT,
    ->     nome VARCHAR(100) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE cargo (
    -> id_cargo INT PRIMARY KEY AUTO_INCREMENT,
    ->     nome VARCHAR(100) NOT NULL UNIQUE,
    ->     fk_empresa INT,
    ->     CONSTRAINT cFkEmpresa
    ->
Display all 758 possibilities? (y or n)
    -> FOREIGN KEY (fk_empresa)
    ->         REFERENCES empresa (id_empresa)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE cargo_permissao (
    -> id_cargo INT,
    ->     id_permissao INT,
    ->
    ->     PRIMARY KEY (id_cargo, id_permissao),
    ->
    ->     FOREIGN KEY (id_cargo) REFERENCES cargo (id_cargo),
    ->     FOREIGN KEY (id_permissao) REFERENCES permissao (id_permissao)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
```

```
mysql> CREATE TABLE usuario(
    ->     id_usuario INT PRIMARY KEY AUTO_INCREMENT,
    ->     nome VARCHAR(100) NOT NULL,
    ->     email_institucional VARCHAR(60) NOT NULL UNIQUE,
    ->     cpf CHAR(11) NOT NULL UNIQUE,
    ->     senha VARCHAR(255) NOT NULL,
    ->     verificado TINYINT DEFAULT 0,
    ->     fk_empresa INT NOT NULL,
    ->     fk_cargo INT DEFAULT 1,
    ->     CONSTRAINT cFkUsuarioEmpresa
    ->         FOREIGN KEY (fk_empresa)
    ->         REFERENCES empresa(id_empresa),
    -> CONSTRAINT cFkUsuarioCargo
    ->
Display all 758 possibilities? (y or n)
    -> FOREIGN KEY (fk_cargo)
    ->         REFERENCES cargo(id_cargo)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE verificacao_email (
    ->     id_verificacao INT PRIMARY KEY AUTO_INCREMENT,
    ->     token VARCHAR(255) NOT NULL UNIQUE,
    ->     dt_expiracao DATETIME NOT NULL,
    ->     fk_usuario INT NOT NULL,
    ->     CONSTRAINT fk_verificacao_usuario
    ->         FOREIGN KEY (fk_usuario)
    ->         REFERENCES usuario(id_usuario)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE localizacao (
    ->     id_localizacao INT PRIMARY KEY AUTO_INCREMENT,
    ->     nome VARCHAR(100) NOT NULL,
    ->     estado VARCHAR(100) NOT NULL,
    ->     cidade VARCHAR(100) NOT NULL,
    ->     cep char(8) NOT NULL
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
```

```
mysql> CREATE TABLE mainframe (
    ->     id_mainframe INT PRIMARY KEY AUTO_INCREMENT,
    ->     hostname VARCHAR(100) NOT NULL UNIQUE,
    ->     modelo VARCHAR(100) NOT NULL,
    ->     numero_serie CHAR(6) NOT NULL,
    ->     status VARCHAR(20) NOT NULL,
    ->     sis_operacional VARCHAR(100) NOT NULL,
    ->     versao_so VARCHAR(20) NOT NULL,
    ->     fk_empresa INT NOT NULL,
    ->     fk_localizacao INT NOT NULL,
    ->     CONSTRAINT ckMainframeStatus
    ->         CHECK (status IN ('ativo', 'inativo', 'manuteno')),
    ->     CONSTRAINT cFkMainframeUsuario
    ->         FOREIGN KEY (fk_empresa)
    ->         REFERENCES empresa(id_empresa),
    ->     CONSTRAINT cFkMainframeLocalizacao
    ->         FOREIGN KEY (fk_localizacao)
    ->         REFERENCES localizacao(id_localizacao)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE fabricante (
    -> id_fabricante INT PRIMARY KEY AUTO_INCREMENT,
    ->     nome_fabricante VARCHAR(45) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE componente (
    -> id_componente INT PRIMARY KEY AUTO_INCREMENT,
    ->     tipo VARCHAR(100) NOT NULL,
    ->     capacidade INT NOT NULL,
    ->     fk_fabricante INT NOT NULL,
    ->     CONSTRAINT cFkComponenteFabricante
    ->
Display all 758 possibilities? (y or n)
    -> FOREIGN KEY (fk_fabricante)
    ->         REFERENCES fabricante(id_fabricante)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
```

```
    -> id_mainframe INT NOT NULL,
    ->     id_componente INT NOT NULL,
    ->     pico_min INT,
    -> pico_max INT,
    ->     percentual INT,
    ->
    ->     PRIMARY KEY (id_mainframe ,id_componente),
    ->
    -> CONSTRAINT cFkParametroMainframe
    ->         FOREIGN KEY (id_mainframe)
    ->         REFERENCES mainframe(id_mainframe),
    ->     CONSTRAINT cFkParametroComponente
    ->         FOREIGN KEY (id_componente)
    ->         REFERENCES componente(id_componente)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO empresa (razao_social, cnpj, dominio)
    -> VALUES
    -> ('Tech Solutions LTDA', '12345678000101', 'techsolutions.com.br'),
    -> ('So Paulo Tech School', '99765432000199', 'sptech.school'),
    -> ('Nexus Sistemas LTDA', '98765432000199', 'nexussistemas.com.br'),
    -> ('Alpha Digital LTDA', '45678912000155', 'alphadigital.com.br');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO cargo (nome, fk_empresa)
    -> VALUES
    -> ('Estagirio', null),
    -> ('Administrador', null);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO permissao (nome)
    -> VALUES
    -> ('Criar'),
    -> ('Visualizar'),
    -> ('Editar'),
    -> ('Excluir');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO cargo_permissao (id_cargo, id_permissao)
    -> VALUES
    -> (1, 2);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO cargo_permissao (id_cargo, id_permissao)
    -> VALUES
    -> (2, 1),
    -> (2, 2),
    -> (2, 3),
    -> (2, 4);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO usuario
    -> (nome, email_institucional, cpf, senha, verificado, fk_empresa, fk_cargo)
    -> VALUES
    -> (
    ->     'Carlos Almeida',
    ->     'carlos.almeida@techsolutions.com.br',
    ->     '12345678901',
    ->     '$2b$10$F5URwsnCnQc5TIv9oWweheTzawJUh8u3FgoxMUV3soPzV1Jl23Dta',
    ->     1,
    ->     1,
    ->     1
    -> ),
    -> (
    ->     'Mariana Santos',
    ->     'mariana.santos@techsolutions.com.br',
    ->     '23456789012',
    ->     '$2b$10$F5URwsnCnQc5TIv9oWweheTzawJUh8u3FgoxMUV3soPzV1Jl23Dta',
    ->     1,
    ->     1,
    ->     2
    -> ),
    -> (
    ->     'Ana Souza',
    ->     'ana.souza@sptech.school',
    ->     '45678901234',
    ->     '$2b$10$F5URwsnCnQc5TIv9oWweheTzawJUh8u3FgoxMUV3soPzV1Jl23Dta',
    ->     1,
    ->     2,
```

```
    -> (nome, estado, cidade, cep)
    -> VALUES
    -> (
    ->     'Data Center So Paulo',
    ->     'So Paulo',
    ->     'So Paulo',
    ->     '01310100'
    -> ),
    -> (
    ->     'Data Center Osasco',
    ->     'So Paulo',
    ->     'Osasco',
    ->     '06016020'
    -> ),
    -> (
    ->     'Data Center Campinas',
    ->     'So Paulo',
    ->     'Campinas',
    ->     '13010000'
    -> ),
    -> (
    ->     'Data Center Rio de Janeiro',
    ->     'Rio de Janeiro',
    ->     'Rio de Janeiro',
    ->     '20040002'
    -> ),
    -> (
    ->     'Data Center Curitiba',
    ->     'Paran',
    ->     'Curitiba',
    ->     '80010000'
    -> ),
    -> (
    ->     'Data Center Belo Horizonte',
    ->     'Minas Gerais',
    ->     'Belo Horizonte',
    ->     '30130000'
    -> );
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
```

```
mysql> INSERT INTO mainframe
    -> (
    ->     hostname,
    ->     modelo,
    ->     numero_serie,
    ->     status,
    ->     sis_operacional,
    ->     versao_so,
    ->     fk_empresa,
    ->     fk_localizacao
    -> )
    -> VALUES
    -> (
    ->     'MF-TECH-001',
    ->     'z16',
    ->     'IBM001',
    ->     'ativo',
    ->     'Linux',
    ->     '9.2',
    ->     1,
    ->     1
    -> ),
    -> (
    ->     'MF-TECH-002',
    ->     'PowerEdge R760',
    ->     'DEL001',
    ->     'ativo',
    ->     'Windows Server',
    ->     '2022',
    ->     1,
    ->     2
    -> ),
    -> (
    ->     'MF-SPTECH-001',
    ->     'z15',
    ->     'IBM002',
    ->     'ativo',
    ->     'Linux',
    ->     '8.8',
    ->     2,
    ->     1
    -> ),
```

```
mysql> INSERT INTO fabricante (nome_fabricante)
    -> VALUES
    -> ('Intel'),
    -> ('AMD'),
    -> ('NVIDIA'),
    -> ('Samsung'),
    -> ('Kingston'),
    -> ('Western Digital');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> INSERT INTO componente
    -> (tipo, capacidade, fk_fabricante)
    -> VALUES
    -> ('CPU', 16, 1),
    -> ('CPU', 32, 2),
    -> ('RAM', 64, 4),
    -> ('RAM', 128, 5),
    -> ('DISCO', 1000, 6),
    -> ('DISCO', 2000, 4),
    -> ('SWAP', 24, 3),
    -> ('SWAP', 48, 3),
    -> ('DISCO', 4000, 6);
Query OK, 9 rows affected (0.01 sec)
Records: 9  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO parametro
    -> (
    ->     id_mainframe,
    ->     id_componente,
    ->     pico_min,
    ->     pico_max,
    ->     percentual
    -> )
    -> VALUES
    ->
    -> -- Mainframe 1
    -> (1, 1, 10, 100, 80),
    -> (1, 3, 20, 90, 75),
    -> (1, 5, 30, 95, 85),
    ->
    -> -- Mainframe 2
    -> (2, 2, 15, 100, 80),
    -> (2, 4, 20, 90, 75),
    -> (2, 6, 30, 95, 85),
    ->
    -> -- Mainframe 3
    -> (3, 1, 10, 100, 80),
    -> (3, 4, 20, 90, 80),
    ->
    -> -- Mainframe 4
    -> (4, 2, 15, 100, 80),
    -> (4, 5, 30, 95, 85),
    ->
    -> -- Mainframe 5
    -> (5, 2, 20, 100, 85),
    -> (5, 3, 20, 90, 75),
    -> (5, 7, 30, 95, 90),
    ->
    -> -- Mainframe 6
    -> (6, 1, 10, 100, 80),
    -> (6, 6, 30, 95, 85),
    ->
    -> -- Mainframe 7
    -> (7, 2, 20, 100, 85),
    -> (7, 4, 20, 90, 80),
    -> (7, 8, 30, 95, 90),
    ->
```

6. Conferindo informações do Banco de Dados

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| noctua_bd          |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.01 sec)

mysql> SHOW TABLES;
+---------------------+
| Tables_in_noctua_bd |
+---------------------+
| cargo               |
| cargo_permissao     |
| componente          |
| empresa             |
| fabricante          |
| localizacao         |
| mainframe           |
| parametro           |
| permissao           |
| usuario             |
| verificacao_email   |
+---------------------+
11 rows in set (0.00 sec)

mysql>
```

7. Desligando/Parando um Container

```
mysql> exit
Bye
bash-5.1# exit
exit
ubuntu@ip-172-31-46-171:~$ docker stop db
db
ubuntu@ip-172-31-46-171:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                      PORTS     NAMES
05a9df2f23c4   mysql:8.0.44   "docker-entrypoint.s…"   13 minutes ago   Exited (0) 5 seconds ago              db
ubuntu@ip-172-31-46-171:~$
```

8. Deletando um Container

```
ubuntu@ip-172-31-46-171:~$ docker rm db
db
ubuntu@ip-172-31-46-171:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-46-171:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-46-171:~$
```

9. Deletando uma Imagem

